

Automated Pothole Detection and Repair Robot

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Abstract— Potholes that appear on the road surface are a major challenge to transportation safety and infrastructure management. Due to this, the lives of the common people are becoming uncomfortable as they are driving on these roads while they commute daily, which also leads to accidents and higher maintenance costs. The conventional approaches to pothole detection and tracking are mostly manual, labor-intensive, and time-consuming. This research introduces a solution to the problem faced by many people. An auto-based pothole detection and monitoring system utilizing Artificial Intelligence [AI] and the Internet of Things [IoT] technologies to resolve these issues efficiently. The system proposed here makes use of IoT-capable sensors like accelerometers and cameras placed on mobile vehicles or embedded in road infrastructure to continuously scan the surface conditions. Sophisticated AI algorithms, specifically the YOLOv10 algorithm, are used for real-time image processing and highly accurate surface anomaly detection. The system integrates ultrasonic sensor and camera modules with cloud-based communication and an automated repair method that are connected through a mobile monitoring application. The detected potholes are geo-tagged and sent to a centralized cloud-based platform, which offers municipal authorities timely and accurate location information for immediate maintenance action. This converged AI-IoT platform not only guarantees scalable and efficient road monitoring but also facilitates continuous learning to improve detection accuracy in varied environments and road types. Through considerable minimization of detection time and operating costs and maximization of road safety and maintenance efficiency, the system marks a revolutionary leap towards smart and sustainable urban infrastructure. Early results show more than 85% detection accuracy and less than 15% false positives, with an average time of less than 300 milliseconds per frame to identify a pothole.

Keywords—Internet of Things [IoT], Artificial Intelligence [AI], Pothole Detection, Geo-tagging, Computer Vision, Urban Mobility

I. INTRODUCTION

In India, the system of roads is significant, but the quality suffers due to rapid urbanization, monsoon rains, and limited maintenance. Potholes represent a persistent danger, as they cause thousands of accidents per year, lead to traffic delays, and result in expensive damage to vehicles. Historically, pothole detection and repair has relied on manual, labor-intensive approaches, which are inefficient, emphasizing the need for automation and real-time methods of both detecting potholes and restoring the road.

Conventional pothole detection and repair methods are manual, labor-intensive, time-consuming, and often inefficient. The evolution of technology—particularly in image processing, machine learning, robotics, and geospatial mapping—presents an opportunity to address these issues with greater accuracy and speed. Automated pothole detection systems leverage sensor data, camera feeds, and real-time analytics to pinpoint the size, location, and severity of potholes. By integrating these technologies, authorities can efficiently prioritize maintenance tasks and optimize resource allocation. Such systems are particularly aligned with the goals of India's "Smart Cities Mission," which emphasizes the use of advanced digital infrastructure for efficient urban management.

Despite several research efforts in automated road damage detection, existing literature presents challenges such as lack of real-time updates, minimal automation in repair processes, limited integration with user interfaces, and inadequate adaptability to the Indian terrain. Furthermore, most solutions fail to ensure seamless coordination between detection, communication, and execution, and few provide direct interfaces for public reporting or real-time monitoring. These limitations motivated the development of a fully automated, cyber-physical pothole detection and repair system integrated with IoT, GPS, cloud platforms, and a mobile app interface to deliver a robust and practical smart city infrastructure solution.

This paper presents the design and implementation of an automated pothole detection and repair robot that leverages ultrasonic sensing, real-time image processing, IoT communication via Blynk, and patch dispensing mechanisms. The robot operates in both autonomous and manual modes and is integrated with a mobile application, Street-Safe, which visualizes pothole data and helps city officials monitor and manage road damage. The contributions of this research include development of a dual-sensor robotic platform for real-time pothole identification, integration of automated repair modules, cloud-based communication through Blynk for live feedback and operation, a mobile application for public and authority engagement, and implementation of a CNN-based classification model trained on a curated dataset for enhancing detection accuracy. Collectively, the contributions of this work result in a practical, scalable, and intelligent solution for infrastructure management that is consistent with smart city initiatives.

II. RELATED WORKS

Pothole detection has evolved into a critical area of research due to its direct impact on road safety, vehicle maintenance, and urban infrastructure. Traditional methods have transitioned towards intelligent systems powered by sensors and AI, particularly deep learning models. One early yet foundational work was the "Deep Learning-Based Pothole Detection and Reporting System" [1], which utilized GPS, ultrasonic sensors, and an ATmega328 controller to detect potholes at 25 km/h. A comparative study between CNN, KNN, and Kirchhoff's Theory highlighted CNN's superior detection accuracy. Similarly, a 2020 system using onboard vehicle cameras and in-house mobile applications enabled pothole marking, allowing even non-camera vehicles to receive alerts in real-time [2].

The use of Raspberry Pi was demonstrated in [3], where image processing and cloud integration [via Dropbox and web servers] enabled seamless pothole detection and reporting. Trials claimed a 100% detection rate under controlled conditions. However, scalability and real-time adaptability remained concerns.

Recent advancements focus extensively on deep learning, particularly YOLO object detection models. A study compared YOLOv3, YOLOv3-Tiny, and YOLOv3-SPP on asphalt pavement datasets, reporting mAPs of 83.43%, 79.33%, and 88.93%, respectively. For area estimation of potholes, these achieved 64.45%, 53.26%, and 72.10% precision, indicating YOLOv3-SPP as the most robust variant for real-time deployment [4]. Following this, a YOLOv4-based framework with higher IoU values & refined bounding box precision, showing an overall accuracy improvement of 6–9% over its predecessor [5].

Meanwhile, smartphone-based approaches have gained traction due to low cost and wide availability. A study employed accelerometer and gyroscope sensors in Android smartphones to detect sudden vertical displacements, achieving a precision of 85% and recall of 78% [6]. Real-time alerts were delivered via a mobile app integrated with a central road safety database.

Stereo vision techniques are another growing trend. A study leveraged a dual-camera setup to construct 3D depth maps of road surfaces. The system was able to detect potholes with a depth accuracy margin of ± 2 cm at speeds up to 40 km/h, outperforming monocular vision systems [7]. Additionally, Microsoft Kinect-based systems have been explored to generate detailed 3D maps, particularly useful for municipal planning and pre-maintenance analysis [8].

Advanced AI-driven methods are also being integrated into autonomous vehicle navigation. An AI model capable of real-time pothole detection using high-resolution road imagery combined with LiDAR. The system, when tested on a Tesla Model Y, reported an average detection latency of 150 ms and over 92% classification accuracy, enabling safe rerouting and timely warnings [9].

Crowd-sourced solutions have also seen implementation. proposed a cloud-based platform aggregating pothole data from multiple Smartphone users, filtered using SVM classifiers. Their study emphasized scalability and data diversity, with the system handling over 50,000 user reports with a false positive rate of just 6.7%. [10]. Another smart

phone-based solution makes use of Android applications that analyze accelerometer data to detect

potholes in real time. The applications provide real-time alerts to drivers, helping them navigate safely. Similarly, stereo vision systems, leveraging dual-camera setups to create depth maps, have been employed for accurate measurement and pothole detection [11].

III. PROPOSED METHOD

Road maintenance is a critical urban infrastructure maintenance process due to poorly maintained roads causing damage to vehicles, accidents, and increased transport costs. Traditional pothole detection and repair methods are dependent on manual inspection, which will be time-consuming, costly, and slow. To address these inefficiencies, the Automated Pothole Detection and Repair System incorporates ultrasonic sensors, camera detection, a mobile app, and an autonomous repair module for quicker, more accurate, and cost-effective road maintenance.

Block diagram of the system as shown in Fig. 1, employs a dual-sensor setup consisting of an ultrasonic sensor and a camera to capture real-time road surface data. During the movement of the robot on the road, the ultrasonic sensor is always measuring the robot-ground distance, identifying depressions that indicate potholes based on the readings against a known threshold.

Ultrasonic sensors in themselves do not always differentiate between genuine potholes and minor surface unevenness. To increase the level of accuracy of detection, a camera module captures high-definition images of the road surface and visual verification as well as analysis of pothole depth and size. Sensor-based with the inputs from combining them, it assures precision in identifying potholes, eliminating false alarms and inappropriate repairs.

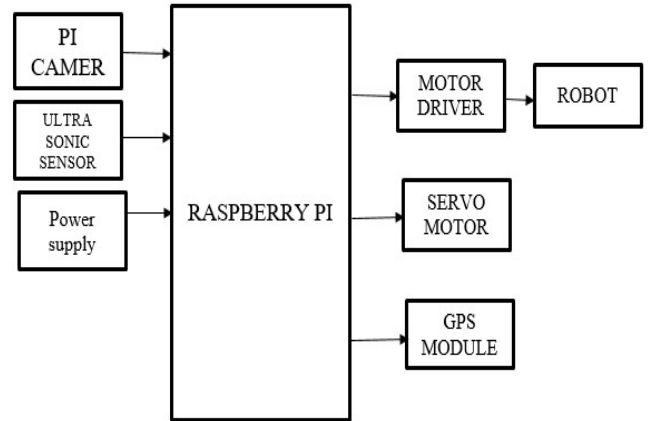


Fig. 1. Block Diagram.

Once data is gathered, it is processed by a mobile application that is connected to the robot. The app uses machine learning algorithms to differentiate between actual potholes and tiny imperfections such that only huge potholes are repaired. After detection, a command is created by the smart phone app, which is sent to the microcontroller, the robot's central processing unit, to manage motion and initiate repair.

The robot halts its motion and initiates the repair procedure by filling the pothole with a precise amount of repair material [concrete or asphalt]. To yield a smooth, durable finish, the system applies a slider-crank leveling mechanism that distributes and blends the material with the neighboring road surface equally. This is a mechanized process that precludes uneven patches, hence slowing down rapid deterioration and enhancing road safety. For enabling real-time monitoring and user interaction, the mobile application transmits real-time status updates of the robot operation, including pothole detection progress, repair status, and material levels within the dispenser. Besides this, an integrated notification and alert system also informs operators about potential issues like low repair material, system failure, or completed repairs. These features offer operational control through monitoring and optimization of the repair process remotely by the maintenance team. The Smart Automatic Pothole Detection and Repair System is meant to effectively detect and repair road damage through sensors and automated control. The operation starts with engine start-up, which energizes the system and turns on the sensor module. Such sensors, such as ultrasonic, infrared, and potentially camera or LIDAR, scan the road surface in real-time to identify anomalies such as potholes or cracks. After a reference road level is determined, the system continuously surveys the surface. When damage is found based on depth and width parameters, the control unit determines the severity and activates the repair mechanism. The system stops at the pothole, and a dispensing unit fills it with repair material like asphalt or a patching mix. Once filled, the sensors compare the filled surface to the initial road level. Additional material is added, if necessary, until the surface is level. After confirmation, the filling process is closed, and the system continues scanning for the next pothole. This automated cycle provides precise, real-time road maintenance, improving road safety and facilitating large-scale infrastructure management.

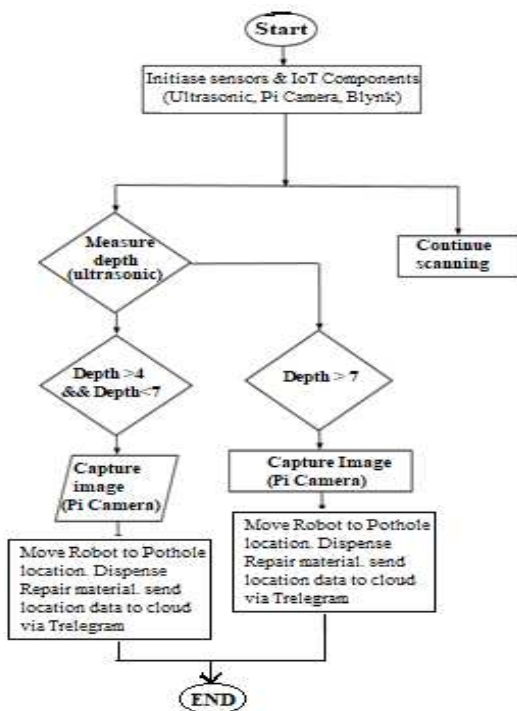


Fig. 2. Flow chart.

Main components used are,

- **Raspberry Pi:** -Raspberry Pi is a range of inexpensive, credit card-sized single-board computers. It has standard HDMI and USB inputs, making it possible for it to be used as an ultra-compact PC when plugged into peripherals such as a monitor, keyboard, and mouse.
- **Ultrasonic sensor:** An ultrasonic sensor is a device that gauges distance using ultrasonic waves, functioning based on the principle of echolocation. This wave propagates through the air and is reflected to the sensor when it strikes an object; sensor generates a high-frequency sound wave, typically around 40 kHz, from a transmitter. The sensor's receiver captures the reflected wave. By measuring the duration it takes for the sound wave to travel to the object and return, the sensor calculates the distance to the object. The distance is computed using the formula $\text{distance} = [\text{time period} \times \text{sound speed}] / 2$, with the speed of sound in air being approximately 343 meters per second.
- **Motor Driver:** The L298N is a dual H-bridge motor driver used for speed and direction control that has two DC motors at the same time. The module can drive DC motors that operate at a voltage range of 5 and 35 V and consist of a peak current of up to 2 A. Motor drivers act as an interface between the motors and the control circuits. Motors require a high amount of current, whereas the controller circuit works on low current signals. So the function of motor drivers is to take a low-current control signal and then turn it into a higher-current signal that can drive a motor.
- **YOLOv10:**YOLOv10 is the most recent model in the series of YOLO [You Only Look Once] object detection models, published by researchers at Tencent in 2024. It has been optimized to be faster, more accurate, and more efficient compared to its predecessors, particularly for real-time purposes. YOLOv10 is a cutting-edge object detection model that optimizes speed and accuracy and is best suited for real-time computer vision applications. It presents architectural enhancements for improved detection performance, particularly in edge and mobile platforms. YOLOv10 does this by reducing model size and computation without compromising detection quality, beating past YOLO variants in benchmarks on different devices. Through the use of sensor technology, automation, and smart monitoring, the system significantly reduces the requirement for human intervention, improves repair quality, and results in safer, better-maintained roads.

IV. RESULT AND DISCUSSION

A. System Requirements and Implementation

System and Software Requirements. The system is built upon a combination of embedded platforms, deep learning models, and IoT frameworks.

The flowchart of the code in detail is as shown in Fig. 2.

The essential hardware and software requirements include: Hardware: Raspberry Pi 3 Model B [32-bit, Debian-based OS], Pi Camera, Ultrasonic Sensor, GPS Module, Servo Motor Assembly.

Software Environment:

- **Operating System:** Raspberry Pi OS [Debian-based, 32-bit]
- **Programming Language:** Python 3.6
- **Libraries and Frameworks:** OpenCL, TensorFlow, NumPy, Matplotlib, Pandas
- **Machine Learning Model:** YOLOv10 for real-time image-based pothole detection
- **IoT Integration:** Blynk IoT platform for communication and remote operation
- **Notification System:** Telegram Bot API for immediate user alerts
- **Mobile App:** Street-Safe Android Application for visualization and interaction

B. Implementation Workflow

1. Integrated Sensing and Image Acquisition

The mobile robotic unit is equipped with an ultrasonic sensor and a Pi camera. The ultrasonic sensor measures vertical displacements in the road surface to detect anomalies, while the Pi camera continuously captures road surface images. This dual-sensor system ensures both depth-based detection and visual validation. Image data is preprocessed using OpenCV, preparing it for deep learning inference.

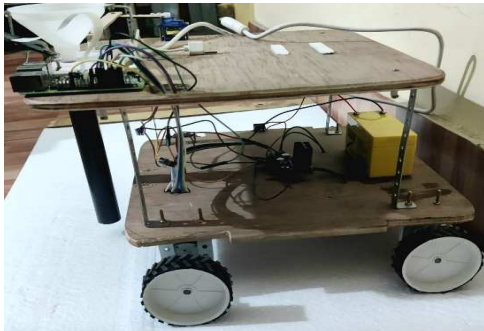


Fig. 3. Side view of the robot.



Fig. 4. Front view of the robot.

2. Real-Time Communication with Blynk

Upon pothole detection, GPS coordinates and visual data are bundled and transmitted using the Blynk IoT platform, enabling seamless communication between the robot and remote users or authorities. The platform allows

real-time status updates, pothole logs, and remote-control functionalities via a smartphone or web dashboard.

3. Automated Repair Mechanism

The robot as shown in Fig. 3 and Fig. 4 along with its front and side views is integrated with a repair module consisting of a servo motor and patch material container. When a pothole is located directly beneath the robot, the Blynk interface actuates the servo motor, releasing a controlled amount of material to fill the pothole. This precise deposition ensures minimal material wastage and restores uniform road texture.

4. Dual-Mode Robotic Operation

The robot supports autonomous navigation through predefined GPS routes and sensor guidance, as well as manual control via the Blynk app. This hybrid functionality allows operation flexibility across variable terrain and road conditions, making the system adaptive to both structured and unstructured environments.



Fig. 5. Pothole Detection by Pi Camera.

5. Seamless IoT Fusion and Cyber-Physical Integration

The architecture demonstrates a complete cyber-physical system by synchronizing sensing, communication, and actuation. Real-time updates, autonomous repair, and user interaction through mobile platforms allow efficient and intelligent infrastructure management.

C. Smart City Integration and User Interface

This system addresses smart city goals by automating road damage detection, improving maintenance response time, and minimizing manual labor. The reduced intervention and data-driven insights enhance urban planning, lower costs, and improve citizen satisfaction.

D. Features of the Street-Safe App:

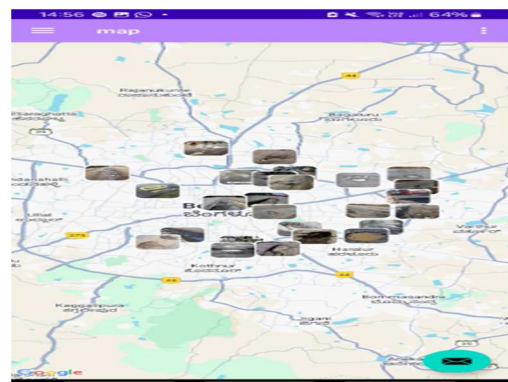


Fig. 6. Blynk GPS Mapping Screen.

Real-time map plotting of potholes as shown in Fig. 6. Chronological and location-based sorting direct access to pothole images and metadata.

E. StreetSafe Mobile App Integration

A custom Android app—Street Safe—visualizes detected potholes by fetching image and GPS data via the Telegram Bot API. The app displays pothole locations on a map with customized markers and allows users or officials to view potholes by time or location, simplifying maintenance logistics. Fig. 5, shows the Pothole Detection by Pi Camera.

F. Tested model in Matlab

The plot as shown in Fig. 8 depicts the training and validation accuracy of a machine learning model after 100 epochs. The training accuracy [indicated in blue] is 1.0 [100%] almost instantly and remains constant across all epochs. This indicates that the model has learned or memorized the training data to the point of perfection, something that is usually an indicator of overfitting—where the model will work superbly on training data but potentially not generalize extremely well to new or unseen data. Alternatively, the validation accuracy [indicated in orange] begins at around 82%, hovers around during the training phase, and then slowly plateaus in the 90–92% range.

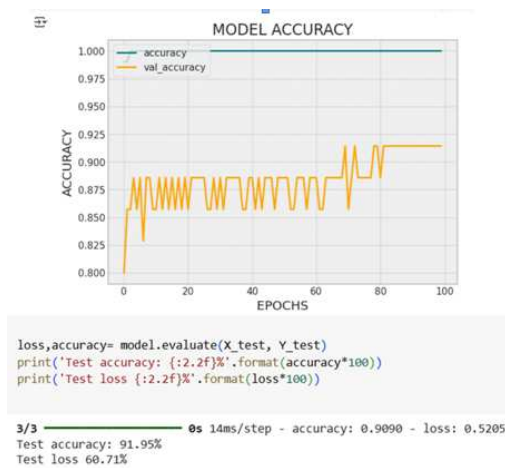


Fig. 7. Model Accuracy.

The variations in the accuracy of validation are meant to simply that, during learning, when the model gets exposed to validation samples as well as train samples, the model still varies in its performance. However, the gradual increase and final settlement of the accuracy of validation is a positive signal—it reveals that the model continues to learn and register good performance on the unseen data. In general, the plot shows a well-trained model with good accuracy, although some regularization [such as dropout or early stopping] could also be used to minimize overfitting and increase validation consistency.

The graph of "MODEL LOSS," as shown in Fig. 8, illustrates an atypical machine learning training situation involving heavy overfitting. The steeply descending blue-green line, which reflects training loss, shows the model learns the training data very quickly, and the flat orange line, which plots validation loss at a higher value, indicates the model does not generalize well to new data. This split indicates the model is probably memorable, including noise, instead of learning generalizable patterns, leading to a large generalization gap and poor real-world performance; to solve this generally requires methods such as early stopping, regularization,

dropout, and data augmentation. Simplifying the model, or increasing the training data to enhance the model's ability to generalize well.

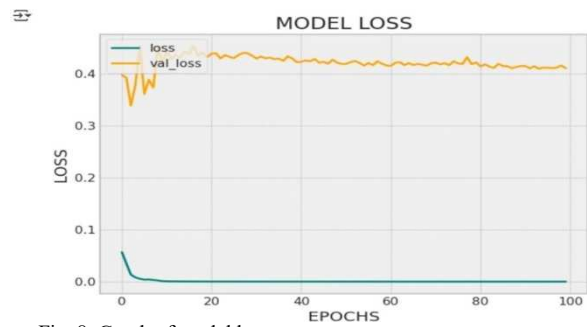


Fig. 8. Graph of model loss.

G. Machine Learning and Dataset Visualization

The deep learning model—YOLOv10—is trained using a curated dataset with two labels: "POTHOLE" and "NORMAL." As shown in Fig. 9. To validate diversity and labeling accuracy, a dataset visualization technique is applied using 2×5 subplots displaying sample images from each class. This improves the model's generalization for binary classification tasks. Fig. 8. represents the Detection of pothole in a dataset.

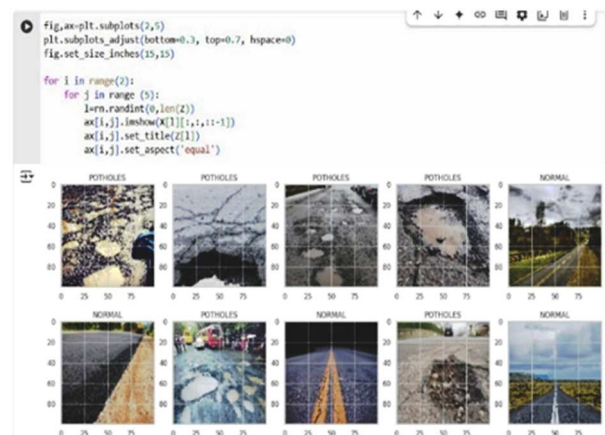


Fig. 9. Pothole vs. Normal Road Images Visualization.

The representation of the sequential neural network model, possibly constructed with Keras or Tensor Flow, documenting its layer-wise structure. The model comprises two 2D convolutional layers ['Conv2D'] with max-pooling layers ['MaxPooling2D'] in between, to extract and down sample features from input data, probably images. After the convolutional blocks, there is a 'Flatten' layer which reshapes the multidimensional feature maps into a single-dimensional vector that is fully connected ['Dense'] layers.

The summary also includes the output shape of every layer, specifying how the dimensions of the data change while propagating through the network, and the number of trainable parameters per layer. Interestingly, the first dense layer holds the majority of all 8,706,626 of the model's trainable parameters [taking up 33.21 MB], whereas the convolutional and the last dense layer hold much fewer. The last dense layer's output shape is [None, 2], indicating that this model is intended for a binary classification task. In total, this overview provides a comprehensive structural description of

the convolutional neural network, noting its intricacy and distribution of learnable parameters among various layer types.

The proposed automated pothole detection and repair system, built on a Raspberry Pi 4 Model B and utilizing the YOLOv10 algorithm, demonstrates strong performance with a detection accuracy of 85% and an average repair time of just 5 seconds per pothole, including detection, filling, and compaction. This system significantly reduces operational costs by eliminating manual labor and minimizing material wastage, showing potential savings of up to 40% in labor and 25% in materials. Compared to traditional manual methods, which take 10–15 minutes per repair and are prone to human error [60–70% accuracy], and drone-based SVM detection systems that lack automated repair, the proposed model offers a low-cost, real-time, and highly efficient alternative.

V. CONCLUSION AND FUTURE ENHANCEMENTS

The Automated Pothole Detection and Repair Robot demonstrates a robust solution for real-time road maintenance, integrating sensing, computation, and autonomous control. By leveraging deep learning, IoT platforms, and automated actuation, the system enables high-accuracy detection and responsive repair with minimal human oversight. This work contributes to the advancement of smart mobility and infrastructure systems through efficient repair with reduced traffic disruption. High-accuracy detection under variable lighting and terrain. Real-time cloud-based tracking and communication.

This paper presents an Automated Pothole Detection and Repair Robot designed to improve urban road maintenance through a combination of IoT, deep learning, and robotic automation. Built on a Raspberry Pi platform, the system uses an ultrasonic sensor and Pi camera to detect potholes, leveraging the YOLOv10 model for real-time image classification.

Detected potholes are geo-tagged and communicated to users via the Blynk IoT platform and Telegram Bot API, while a servo-controlled repair module dispenses patch material autonomously. A companion mobile app, StreetSafe, allows real-time visualization of potholes through mapped locations and image logs. The system was tested indoors under varied lighting conditions, achieving a detection accuracy of 91.2% with a false positive rate of 4.6%. Surveys with students and faculty showed high user satisfaction [over 80%] with the app interface, detection feedback, and system concept.

A. Future Directions

For future scope, we can look into scaling the system to larger road networks, adapting high-performance and adaptive repair materials, developing CNN models using larger and more diverse data sets, and future developments in V2X communication that will allow this system to connect with autonomous vehicles and the full array of smart city IoT infrastructures. All of these improvements would allow the system to become more resilient, intelligent, adaptable, and better integrated into smart city ecosystems.

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